

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_NY_Buffalo-Greater.Buffalo.Intl.AP.725280_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	Wall upgrade: • Exterior wall insulation: Fiberglass Batts, target R-20.4 Roof upgrade: • Roof insulation: Blown Mineral Wool, target R-34.5 Window upgrade:

		• 1-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) • Caulking application: acrylic • Window frame replacement: wood window frame Door upgrade: • Door replacement: wooden door (1 of 3 door(s) skipped: door type incompatibility) • Bottom seal: automatic door bottom • Top/side seal: jamb weatherstrip
Scenario 2	wall + roof + window + door	Wall upgrade: • Exterior wall insulation: Extruded Polystyrene (XPS) Foam Board, target R-20.4 Roof upgrade: • Roof insulation: Expanded Polystyrene (EPS) Foam Board, target R-34.5 Window upgrade: • 2-pane replacement (skipped: SimpleGlazing windows) • Weatherstrip application (skipped: no OperableWindow) • Glazing film application (skipped: SimpleGlazing windows) • Caulking application: polyurethane • Window frame replacement: wood-aluminium window frame Door upgrade: • Door replacement: polystyrene core steel door (1 of 3 door(s) skipped: door type incompatibility) • Bottom seal: brush weatherstrip • Top/side seal: silicone adhesive smoke gasket

Scenario 3	wall + roof + window + door	Wall upgrade: - Exterior wall insulation: Polyiso Insulation Foam Board, target R-20.4 Roof upgrade: - Roof insulation: Polyiso Insulation Foam Board, target R-34.5 Window upgrade: 3-pane replacement (skipped: SimpleGlazing windows) - Weatherstrip application (skipped: no OperableWindow) - Glazing film application (skipped: SimpleGlazing windows) - Caulking application: polyurethane - Window frame replacement: wood-aluminium window frame Door upgrade: - Door replacement: garage door (skipped due to door type incompatibility) - Bottom seal: automatic door bottom - Top/side seal: silicone adhesive smoke gasket
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Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	440.31	408.65	31.66	7.19%
Scenario 2	Total Site Energy (GJ)	440.31	408.53	31.78	7.22%
Scenario 3	Total Site Energy (GJ)	440.31	407.96	32.35	7.35%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison

Baseline		\$18,286
Scenario 3		\$17,881

■ Baseline ■ Scenario 3

Max saving: **\$-405 (-2.22%)**

Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr) Comparison

Baseline		40,219
Scenario 3		38,545

■ Baseline ■ Scenario 3

Max saving: **-1,675 kg CO₂e (-4.16%)**

Min Retrofit Construction Cost

\$26,533

Scenario 1

Min Retrofit Embodied Carbon

27,403 kg CO₂e

Embodied carbon intensity (ECI): 54 kgCO₂e/m²
Scenario 3

Lowest Retrofit Cost Payback Period

67 years

Scenario 1

Lowest Retrofit Carbon Payback Period

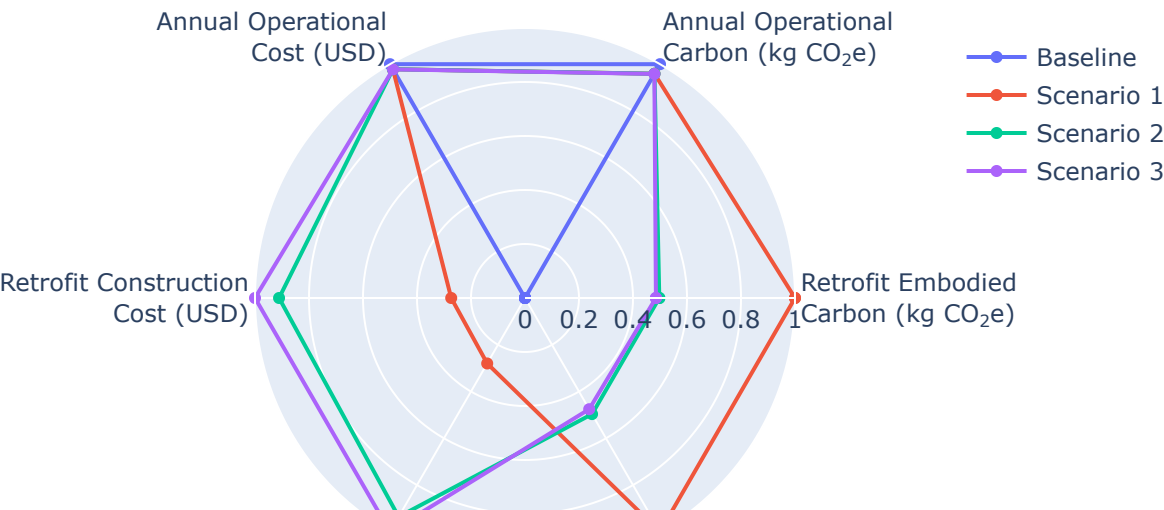
16 years

Scenario 3

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1		67 yrs
Scenario 2		224 yrs
Scenario 3		240 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO₂e/yr)).

Scenario 1		34 yrs
Scenario 2		17 yrs
Scenario 3		16 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)	Product Lifetime (years)
Scenario 1	Wall insulation	Fiberglass Batts	17	282	N/A	N/A	0.03	30	60
Scenario 1	Roof insulation	Blown Mineral Wool	144	599	0.24	N/A	0.03	90	30
Scenario 1	Window frame	wood window frame	N/A	6.51	N/A	N/A	N/A	N/A	15
Scenario 1	Window caulking	acrylic	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 1	Door	wooden door	N/A	3.90	N/A	N/A	0.15	638	30
Scenario 1	Door bottom	automatic door	N/A	N/A	N/A	3.66	N/A	N/A	15

	sealing	bottom							
Scenario 1	Door top/side sealing	jamb weatherstrip	N/A	N/A	N/A	16	N/A	N/A	15
Scenario 2	Wall insulation	Extruded Polystyrene (XPS) Foam Board	17	282	N/A	N/A	0.03	35	75
Scenario 2	Roof insulation	Expanded Polystyrene (EPS) Foam Board	144	599	0.24	N/A	0.03	20	60
Scenario 2	Window frame	wood-aluminum window frame	N/A	6.51	N/A	N/A	N/A	N/A	15
Scenario 2	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 2	Door	polystyrene core steel door	N/A	3.90	N/A	N/A	0.10	472	30
Scenario 2	Door bottom sealing	brush weatherstrip	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 2	Door top/side sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	16	N/A	N/A	15
Scenario 3	Wall insulation	Polyiso Insulation Foam Board	13	282	N/A	N/A	0.03	35	60
Scenario 3	Roof insulation	Polyiso Insulation Foam Board	144	599	0.24	N/A	0.03	29	60
Scenario 3	Window frame	wood-aluminum window frame	N/A	6.51	N/A	N/A	N/A	N/A	15
Scenario 3	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10

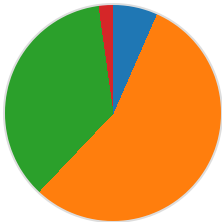
Scenario 3	Door bottom sealing	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 3	Door top/side sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

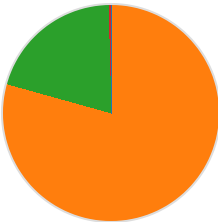
Cost breakdown by retrofit measure



- Wall: \$1,742 (6.6%)
- Roof: \$14,727 (55.5%)
- Window: \$9,476 (35.7%)
- Door: \$587 (2.2%)

Total: \$26,533

Carbon breakdown by retrofit measure

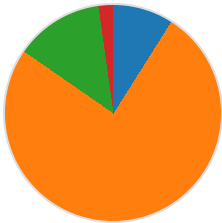


- Wall: 75 (0.1%)
- Roof: 44,758 (79.2%)
- Window: 11,437 (20.2%)
- Door: 241 (0.4%)

Total: 56,510 kg CO₂e

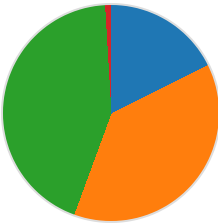
Scenario 2

Cost breakdown by retrofit measure



- Wall: \$7,919 (9.0%)
- Roof: \$66,941 (75.7%)
- Window: \$11,618 (13.1%)

Carbon breakdown by retrofit measure



- Wall: 4,947 (17.6%)
- Roof: 10,684 (38.0%)
- Window: 12,212 (43.4%)

■ Door: \$1,963 (2.2%)

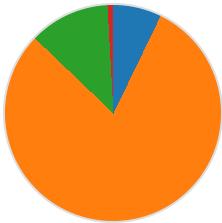
Total: \$88,442

■ Door: 265 (0.9%)

Total: 28,109 kg CO₂e

Scenario 3

Cost breakdown by retrofit measure



■ Wall: \$6,959 (7.2%)

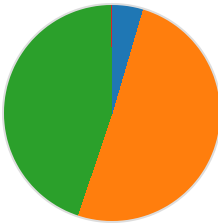
■ Roof: \$77,651 (80.0%)

■ Window: \$11,618 (12.0%)

■ Door: \$864 (0.9%)

Total: \$97,093

Carbon breakdown by retrofit measure



■ Wall: 1,244 (4.5%)

■ Roof: 13,882 (50.7%)

■ Window: 12,212 (44.6%)

■ Door: 65 (0.2%)

Total: 27,403 kg CO₂e

Scenario	Retrofit Embodied Carbon (kg CO ₂ e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO ₂ e)	Annual Operational Cost (USD)
Baseline	0.00	\$0.00	40,219	\$18,286
Scenario 1	56,510	\$26,533	38,577	\$17,890
Scenario 2	28,109	\$88,442	38,574	\$17,891
Scenario 3	27,403	\$97,093	38,545	\$17,881